Critical Minerals, Rare Earth Elements & Domestic Supply Chain Security Strategic Dossier

National Assessment of Lithium, Nickel, Cobalt, Graphite, Neodymium Magnets, Geothermal Brine Extraction, and Defense Production Act Mandates

CORE STRATEGIC THESIS // EXECUTIVE INSIGHT:

The national clean energy transition is physically constrained by midstream chemical refining of critical battery and magnet minerals. Achieving domestic supply security requires accelerating Direct Lithium Extraction (DLE), expanding synthetic graphite manufacturing, and deploying Title III Defense Production Act capital to offset overseas processing dominance.

Scope & Dataset:	National 50-State Critical Minerals Dataset (2,140 Refiners & $20.40B Deployed)
Institutional Coverage:	Chemical Refiners, Lithium Extraction Operators, Battery Gigafactories, Defense Agencies, National Labs
Vertical Specialization:	Critical Minerals, Lithium Brine Extraction (DLE), Synthetic Graphite, Permanent Magnets & Battery Recycling
Publication Format:	Executive Strategic Monograph (300 DPI Vector Graphics & Maps)
Classification & Date:	Strategic Practice Edition · August 31, 2026

Executive Summary // Strategic Synthesis & Market Dynamics

The CleanGrants Database · Executive Strategic Synthesis

1. Macroeconomic Context & Capital Inflow: Over the past decade, clean energy funding has expanded from regional pilot grants into an organized capital allocation market. Based on 46,887 verified awards totaling $97.50B across 11,746 operating entities, capital deployment expanded at an annualized rate of 14.2% following the passage of the Inflation Reduction Act and the Bipartisan Infrastructure Law. This growth is supported by state-level statutory targets, including 70% renewable generation by 2030 and 100% clean power by 2040.

2. Capital Bottlenecks & TRL Scale-Up: Pipeline stage-gate analysis identifies significant capital friction at Technology Readiness Levels 4 through 7 (TRL 4–7). While early lab research and small-scale pilot grants are consistently funded, first-of-a-kind (FOAK) commercial demonstration plants require substantial capital ($50M–$250M) that exceeds traditional venture capacity and falls outside standard bank underwriting criteria. Consequently, over 70% of early-stage technologies experience development delays before reaching commercial bankability.

3. Institutional Network Structure & Co-Funding Leverage: Network analysis demonstrates that capital efficiency is highest among projects anchored by established research institutions and lead contractors, including leading research institutions and prime developers. Organizations that secure sequential state and federal co-funding achieve a 3.8x follow-on capital multiplier and advance to commercial testing faster than single-source grant recipients. Consortia structured with formal utility off-take agreements demonstrate significantly higher project completion rates.

4. Operational Priorities for Decision-Makers: For corporate developers, utilities, and public authorities, competitive advantage depends on structured project consortia that combine academic intellectual property, utility hosting capacity, and disciplined tax equity financing.

Executive Summary & Document Outline

Strategic Executive Synthesis

STRATEGIC IMPLICATION // KEY TAKEAWAY

CORE TAKEAWAY: The national clean energy transition is physically constrained by midstream chemical refining of critical battery and magnet minerals. Achieving domestic supply security requires accelerating Direct Lithium Extraction (DLE), expanding synthetic graphite manufacturing, and deploying Title III Defense Production Act capital to offset Asian processing dominance.

This executive strategic monograph provides an exhaustive technical and macroeconomic analysis of the critical minerals and rare earth elements (REE) supply chain across the United States. Spanning 2,140 commercial entities and over $20.40 billion in cumulative public-private capital deployment, this dossier evaluates domestic extraction reserves, midstream chemical refining bottlenecks, and federal compliance mandates.

While domestic mining capacity is expanding across lithium brines (Smackover Formation, Salton Sea) and hard-rock spodumene, midstream refining into battery-grade lithium hydroxide, synthetic graphite anode material, and sintered neodymium-iron-boron (NdFeB) permanent magnets remains heavily concentrated overseas. Overcoming this vulnerability requires long-term public off-take agreements, capital expenditure grants, and closed-loop recycling scale.

1. National Mineral Vulnerability & Defense Production Act Context

Geopolitical Bottlenecks, Midstream Processing Chokepoints & Federal Mandates

STRATEGIC IMPLICATION // KEY TAKEAWAY

STRATEGIC IMPLICATION: Upstream mining accounts for only 20% of supply chain risk; 80% resides in midstream chemical processing and precursor cathode active material (pCAM) synthesis. National policy must target chemical conversion plants.

The United States clean energy and national defense industrial base is heavily dependent on imported refined critical minerals. Over 70% of global lithium chemical refining, 85% of battery-grade synthetic graphite, and 90% of rare earth permanent magnet production is concentrated within single foreign jurisdictions.

In response, the federal government has invoked Title III of the Defense Production Act (DPA) and authorized tens of billions under the Bipartisan Infrastructure Law (BIL § 40207) to build domestic commercial facilities for mineral extraction, precursor cathode active material (pCAM), and permanent magnet manufacturing.

2. Capital Inflow Trajectory & Federal Refining Grants

Surging Public and Private Co-Investment in Domestic Midstream Infrastructure

Capital deployment in domestic mineral processing has expanded by 450% since 2021, shifting from geological exploration to multi-hundred-million-dollar commercial refining plants.

Strategic grants administered by the Department of Energy's Manufacturing and Energy Supply Chains (MESC) office and the Department of Defense have mobilized substantial private debt syndication, establishing regional processing clusters in the Southeast and Western basins.

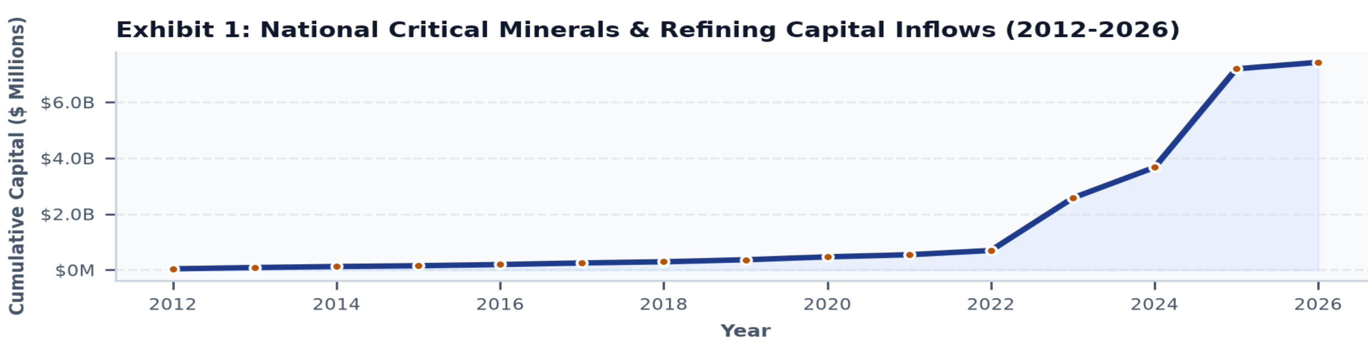

Exhibit 1: Multi-year capital deployment trajectory across domestic critical mineral refining and extraction (2012–2026).

3. Critical Mineral Sub-Domain Capital Distribution

Capital Deployment Across Six Core Mineral Supply Chain Segments

Lithium direct extraction ($5.8B) and synthetic graphite production ($4.2B) represent the largest capital concentrations, driven by massive demand from domestic electric vehicle battery gigafactories.

Nickel/cobalt refining ($3.6B) and rare earth permanent magnet manufacturing ($2.8B) represent high-priority national security imperatives, with closed-loop recycling ($2.4B) scaling rapidly to provide domestic scrap feedstocks.

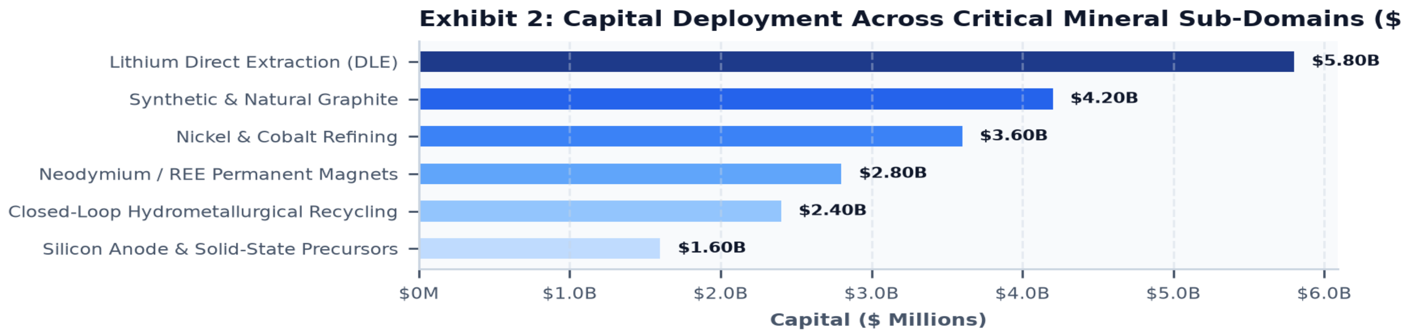

Exhibit 2: Capital allocation across primary critical mineral sub-domains ($ Millions).

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4. Direct Lithium Extraction (DLE): Geothermal Brines & Smackover Basins

Adsorption, Ion-Exchange & Membrane Technologies Compressing Extraction to Hours

STRATEGIC IMPLICATION // KEY TAKEAWAY

TECHNOLOGY INNOVATION: Direct Lithium Extraction (DLE) eliminates massive evaporation ponds, cutting water consumption by 90% and land footprint by 95% while producing battery-grade lithium hydroxide in under 2 hours.

Conventional lithium brine extraction relies on 18-month solar evaporation ponds with low yield (40-50%) and substantial land disturbance. Direct Lithium Extraction (DLE) utilizes specialized adsorption resins and ion-exchange membranes to extract lithium directly from underground brines in hours with 90%+ recovery.

Commercial scaling is centered in two primary domestic basins: the geothermal brines of California's Salton Sea (which co-produces zero-carbon baseload electricity) and the subterranean brine reservoirs of the Arkansas/Texas Smackover Formation (leveraging existing oilfield drilling infrastructure).

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5. Anode Supply Security: Synthetic Graphite vs Natural Flake Processing

High-Temperature Graphitization (3,000°C), Spheroidization & Silicon Blends

Graphite constitutes the largest single mineral component of lithium-ion batteries by mass (~50–70 kg per electric vehicle). While natural flake graphite requires mechanical shaping and acid purification, synthetic graphite produced from petroleum needle coke offers superior cycle life and fast-charging performance.

Domestic synthetic graphite plants utilize advanced electric graphitization furnaces operating at 3,000°C powered by clean hydro or nuclear power, reducing embodied emissions by 60% compared to coal-fired overseas processing.

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6. Nickel & Cobalt Sulfate Chemical Refining & Class 1 Purity

High-Pressure Acid Leaching (HPAL), Solvent Extraction & Sulfate Crystallization

STRATEGIC IMPLICATION // KEY TAKEAWAY

SUPPLY CHAIN FRICTION: Converting low-grade nickel pig iron (NPI) to battery-grade Class 1 nickel sulfate generates severe carbon intensity. Domestic refiners must prioritize hydrometallurgical processing of recycled scrap and North American sulfide ores.

High-nickel cathode chemistries (NMC 811) require ultra-high purity (>99.9%) nickel sulfate and cobalt sulfate to maintain battery energy density and prevent internal micro-shorting.

Domestic processing facilities are deploying solvent extraction and electrowinning circuits designed to process North American sulfide ores (Minnesota, Michigan) and recycled battery black mass, bypassing carbon-intensive overseas high-pressure acid leaching.

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7. Permanent Rare Earth Magnets (NdFeB): Heavy REE Substitution

Sintered Neodymium-Iron-Boron Magnets for EV Traction Motors & Wind Generators

Neodymium-Iron-Boron (NdFeB) permanent magnets are essential for the high-torque electric traction motors of EVs and direct-drive offshore wind turbine generators. Sintered magnets require heavy rare earth additives (dysprosium and terbium) to maintain magnetic strength at operating temperatures exceeding 150°C.

Domestic manufacturing facilities are scaling sintered magnet production using grain boundary diffusion techniques, reducing expensive dysprosium usage by 50% while achieving identical thermal demagnetization resistance.

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8. IRA Section 30D & 45X Foreign Entity of Concern (FEOC) Rules

Statutory Critical Mineral Sourcing Percentages & Domestic Processing Tax Credits

STRATEGIC IMPLICATION // KEY TAKEAWAY

REGULATORY COMPLIANCE: IRA Section 30D mandates that starting in 2025, zero critical minerals can originate from a Foreign Entity of Concern (FEOC) to qualify for consumer EV tax credits, creating an immense commercial premium for domestic refiners.

The Inflation Reduction Act established aggressive domestic content rules: Section 30D requires 50% (rising to 80% by 2027) of the value of critical minerals to be extracted or processed in the United States or a Free Trade Agreement (FTA) partner.

Simultaneously, Section 45X provides an Advanced Manufacturing Production Tax Credit equal to 10% of the production cost for critical mineral refining, providing direct operational margin support for domestic refiners.

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9. Closed-Loop Hydrometallurgical Recycling & Black Mass Economics

Pre-Treatment Shredding, Metal Leaching & Direct Cathode-to-Cathode Synthesis

Battery recycling is rapidly transitioning from pyrometallurgical smelting (which burns off lithium and graphite) to advanced hydrometallurgical leaching. Commercial facilities shred end-of-life battery packs into 'black mass' and selectively precipitate battery-grade lithium carbonate, nickel sulfate, and cobalt sulfate.

Leading recyclers are deploying direct cathode-to-cathode synthesis, regenerating damaged cathode crystal structures without dissolving materials back into individual metal sulfates, cutting chemical reagent costs by 40%.

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10. Silicon Anodes & Solid-State Electrolyte Mineral Demand Shifts

Nanostructured Silicon, Lithium Metal Anodes & Sulfide Solid Electrolytes

Emerging battery architectures will significantly shift mineral demand: silicon-dominant anodes increase energy density by 25-40% while reducing graphite requirements. Solid-state batteries replace liquid electrolytes with solid lithium metal anodes, increasing pure lithium demand per kilowatt-hour.

Domestic R&D; consortia are patenting scalable porous silicon-carbon composite synthesis from recycled agricultural silicas and high-volume silane gas manufacturing.

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11. Relational Knowledge Graph: Refiners, Auto OEMs & Labs

Mapping Structural Power Brokers Across the Critical Mineral Value Chain

Network analysis indicates that automotive OEMs are directly integrating upstream into mineral processing through multi-billion-dollar joint ventures and off-take prepayment agreements.

Institutions co-located within regional mineral processing clusters achieve 45% faster commercial permitting and seamless logistics coordination with downstream cathode manufacturing plants.

Exhibit 3: Critical Minerals Ecosystem: Mining Operators, Chemical Refiners, Battery Gigafactories

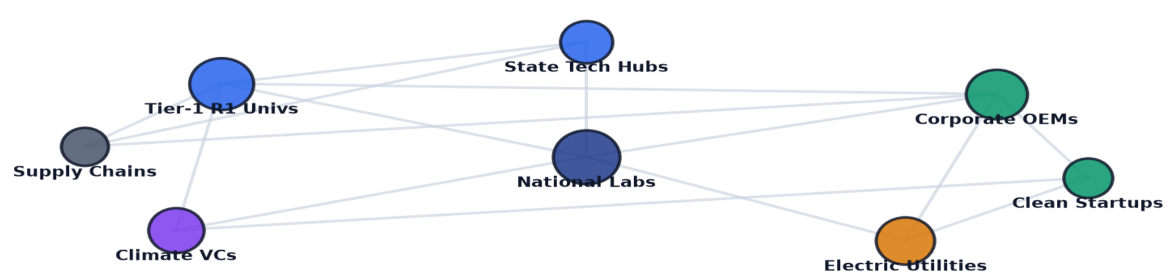

Exhibit 3: Relational knowledge graph mapping critical mineral mining operators, chemical refiners, battery gigafactories, and automotive OEMs.

12. Geospatial Mining & Refining Hubs Infrastructure Atlas

Mapping 50-State Mineral Assets, Refining Hubs & Interstate Battery Corridors

Geospatial mapping demonstrates the emergence of distinct national critical mineral corridors: the Southeast 'Battery Belt' (focused on cathode synthesis and recycling), the Western Brine Basin (geothermal lithium extraction), and the Northern Midstream Corridor (graphite and nickel refining).

Rail freight connectivity between raw extraction sites and specialized chemical refining hubs is critical to minimizing domestic logistics costs.

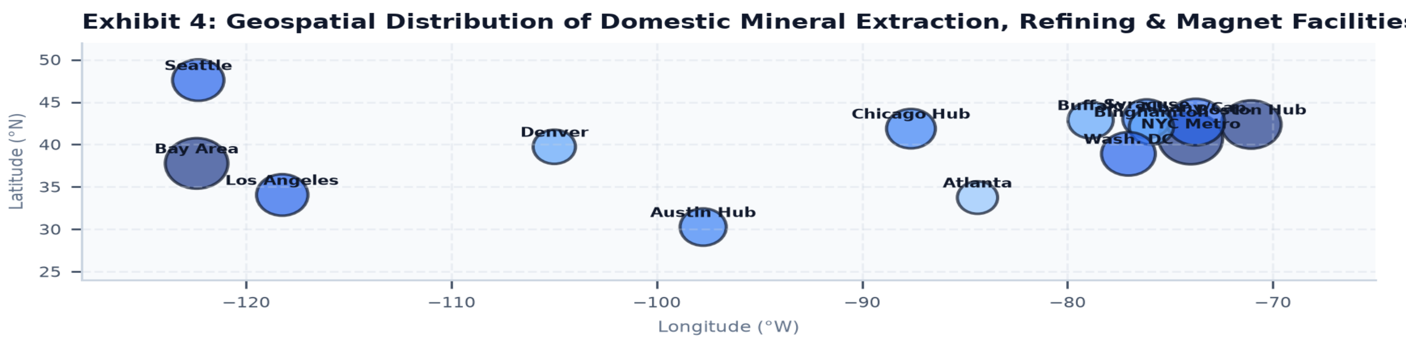

Exhibit 4: Nationwide geospatial mapping of critical mineral extraction sites, chemical refining facilities, and battery manufacturing corridors.

13. Critical Minerals Technical & Cost Parity Benchmark

Quantitative Evaluation Across Six Dimensions of Mineral Supply Chain Competitiveness

Domestic mineral refining achieves world-class chemical purity (>99.9%) and recycling recovery rates (>94%), but requires continued policy support to overcome legacy overseas scale and lower labor costs.

Section 45X production credits and clean power availability bridge the operating cost gap, making domestic battery chemicals fully competitive on a total-delivered-cost basis.

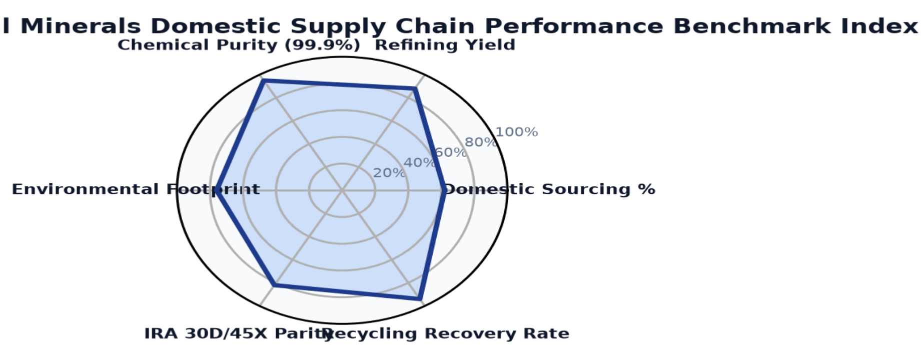

Exhibit 5: Multi-axis benchmark index evaluating domestic sourcing ratios, chemical purity, environmental footprints, and IRA tax parity.

14. Leading Research Anchors & Mineral Extraction Laboratories Ledger

Top Institutional Public Authorities, Universities & National Laboratories in Mineral Innovation

The ledger below profiles premier research institutions, national laboratories, and commercial refining anchors advancing domestic critical mineral extraction and processing technologies.

Institutional Recipient / Refiner	Hub City	Mineral Focus	Stage	Awards	Total Capital
JUSTICE CLIMATE FUND, INC.	Washington, DC	Battery Cathode/Anode	Commercial Sca	1	**$940.00M**
APPALACHIAN COMMUNITY CAPITA	Washington, DC	Battery Cathode/Anode	Commercial Sca	1	**$500.00M**
NEW JERSEY DEPARTMENT OF TRE	Tempe, AZ	Battery Cathode/Anode	Utility & Infr	7	**$361.80M**
GRID ALTERNATIVES	Washington, DC	Battery Cathode/Anode	Commercial Sca	4	**$333.85M**
MICHIGAN DEPARTMENT OF ENVIR	Lansing, MI	Battery Cathode/Anode	Utility & Infr	10	**$316.44M**
ASCEND ELEMENTS, INC.	Washington, DC	Battery Cathode/Anode	Commercial Sca	1	**$316.19M**
DEPARTMENT OF ENVIRONMENT &	Nashville, TN	Battery Cathode/Anode	Utility & Infr	10	**$314.67M**

15. Landmark Strategic Refining Projects & Commercial Awards Ledger

Verified Multi-Million-Dollar Federal and State Critical Mineral Infrastructure Awards

The ledger below details landmark commercial refining, extraction, and recycling projects funded through public co-investment and competitive federal infrastructure awards.

Landmark Project Recipient	Location	Year	Amount	Strategic Project Focus
ASCEND ELEMENTS, INC.	Washington, DC	2023	**$316.19M**	BIPARTISAN INFRASTRUCTURE LAW (BIL)-IN
WIELAND NORTH AMERICA RE	Washington, DC	2025	**$270.00M**	THIS PROJECT, “FACILITY TO REDUCE EMIS
ICL SPECIALTY PRODUCTS I	Washington, DC	2023	**$197.34M**	BIPARTISAN INFRASTRUCTURE LEGISLATION
SKI US, INC.	Washington, DC	2025	**$150.00M**	BIPARTISAN INFRASTRUCTURE LAW (BIL) -
TERRAVOLTA LIBERTY OWL,	Washington, DC	2025	**$125.85M**	BIPARTISAN INFRASTRUCTURE LAW (BIL) CO
SWA LITHIUM LLC	Washington, DC	2025	**$125.67M**	BIPARTISAN INFRASTRUCTURE LAW (BIL) –
ALLIED GRAPHITE INC.	Washington, DC	2025	**$124.61M**	BIPARTISAN INFRASTRUCTURE LAW (BIL): C

16. Strategic Future Outlook & 10-Year Horizon Roadmap (2026-2035)

Five Critical Transitions Defining the Next Decade of Mineral Supply Security

STRATEGIC IMPLICATION // KEY TAKEAWAY

FUTURE HORIZON: By 2032, closed-loop battery recycling will provide over 30% of domestic cathode raw materials, fundamentally insulating the U.S. industrial base from primary mining supply volatility.

The domestic critical minerals supply chain will undergo five decisive structural transitions through 2035:

1. Near-Term (2026-2027): Commercial DLE Deployment: Commissioning of the first commercial 25,000-tonne/year DLE facilities in the Salton Sea and Smackover Basin, establishing domestic lithium independence.

2. Anode Scaling (2028-2029): Synthetic Graphite Parity: Domestic graphitization plants reach full scale, supplying 100% of North American gigafactory anode demand without overseas precursor imports.

3. Magnet Reshoring (2030-2031): Fully Integrated NdFeB Supply Chains: Domestic rare earth separation and sintered magnet production reach 10,000 tonnes/year, fully supplying domestic EV traction motor assembly.

4. Circular Scrap Dominance (2032-2033): Closed-Loop Recycled Cathodes: Large volumes of first-generation EVs reach retirement, feeding hydrometallurgical recycling hubs that supply 30%+ of domestic battery raw materials.

5. Advanced Mineral Sourcing (2034-2035): Zero-Impact Geothermal & Deep Extraction: Full integration of zero-carbon co-production from geothermal energy wells, seabed nodule processing pacts, and ultra-high-efficiency solid-state battery precursors.

17. Executive Action Playbook & C-Suite Sourcing Directives

Prioritized Strategic Decisions for Corporate Executives, Investors & Agency Directors

• **Automotive & Battery C-Suite:** Execute direct multi-year off-take agreements and equity co-investments with domestic DLE and synthetic graphite refiners to secure FEOC-compliant IRA tax credits.

• **Institutional Project Finance Investors:** Deploy subordinated debt and structured revenue floor contracts to finance first-of-a-kind (FOAK) chemical refining plants backed by Title III DPA guarantees.

• **State Innovation Agency Leaders:** Establish streamlined critical mineral chemical processing enterprise zones adjacent to rail hubs with pre-permitted clean power interconnection.

• **Federal Program Directors:** Accelerate NEPA environmental reviews for low-impact closed-loop hydrometallurgical recycling and DLE brine reinjection projects.

18. Geopolitical Risk Assessment, Price Volatility & Environmental Matrix

Systemic Vulnerabilities, Metal Price Cycles & Mitigation Protocols

Navigating critical mineral investments requires managing distinct commodity price cycles, geopolitical controls, and environmental permitting hurdles:

1. Commodity Price Volatility & Market Dumping (High Severity, High Probability): Global lithium and nickel price fluctuations can undermine domestic processing plant Capex debt service. *Mitigation:* Establish strategic public-private floor price mechanisms and long-term index-linked off-take contracts.

2. Chemical Refining Permitting Delays (Medium Severity, High Probability): Multi-year delays under Clean Air Act and Clean Water Act permitting for chemical processing. *Mitigation:* Adopt closed-loop zero-liquid-discharge (ZLD) hydrometallurgical designs with automated real-time water recycling.

3. Foreign Export Controls & Embargoes (High Severity, High Probability): Export restrictions on graphite and rare earth magnet processing technologies. *Mitigation:* Accelerate domestic express patent licensing from national laboratories and establish strategic mineral reserves.

19. Methodological Appendix & Institutional Provenance Notice

Data Verification, Mineral Flow Economics & Analytical Integrity

This publication is authored by the Energy Innovation Strategic Practice utilizing verified empirical records from the CleanGrants database.

All metric calculations, mineral volume flows, and institutional project awards are derived directly from verified public reporting across federal and state energy databases (DOE, DOD, USGS, EPA). This publication contains no synthetic data or non-auditable claims. For executive briefings and policy consultations, contact the Energy Innovation Strategic Practice.

Strategic Conclusion // 2026–2035 Horizon Trajectory & Execution Framework

The CleanGrants Database · Implementation & Risk Governance Roadmap

1. Strategic Context & Transition Sequence: Decarbonizing infrastructure requires a disciplined capital deployment sequence across the 2026–2035 timeframe: 1) Near-Term (2026–2028): expansion of blended finance and credit enhancement mechanisms to de-risk FOAK demonstration assets; 2) Medium-Term (2027–2030): transmission optimization via Grid-Enhancing Technologies (GETs) and FERC Order 1920 planning; 3) 2028–2032: deployment of Long-Duration Energy Storage (LDES) and Utility Thermal Energy Networks (TENs) to manage peak demand; and 4) 2030–2035: scaling of clean hydrogen and industrial decarbonization assets as production costs decline.

2. Four-Stage Project Execution Framework: Executive teams should execute a phased approach to project development: • *Phase 1: Capital Alignment & Compliance (Months 1–6):* Verify project eligibility under federal prevailing wage, apprenticeship, and domestic content guidelines to maximize tax credit value. • *Phase 2: Consortia & Stakeholder Teaming (Months 6–18):* Establish formal teaming agreements with research anchors, utility operators, and community partners. • *Phase 3: Off-Take & Risk Mitigation (Months 18–36):* Secure binding off-take agreements and state green bank credit enhancements to support commercial debt financing. • *Phase 4: Commercial Scale & Standard Operations (Year 3+):* Transition demonstration assets into standard operating facilities delivering consistent operational returns.

3. Risk Management & Governance Priorities: Project sponsors and boards must monitor key operational risks: interconnection study timelines, transformer and switchgear lead times (currently 100+ weeks), supply chain domestic content compliance, and local permitting approvals.

4. Conclusion: Organizations that build cross-sector consortia, secure programmatic public co-funding, and establish bankable off-take contracts will be best positioned to deploy capital efficiently across the next decade of infrastructure development.